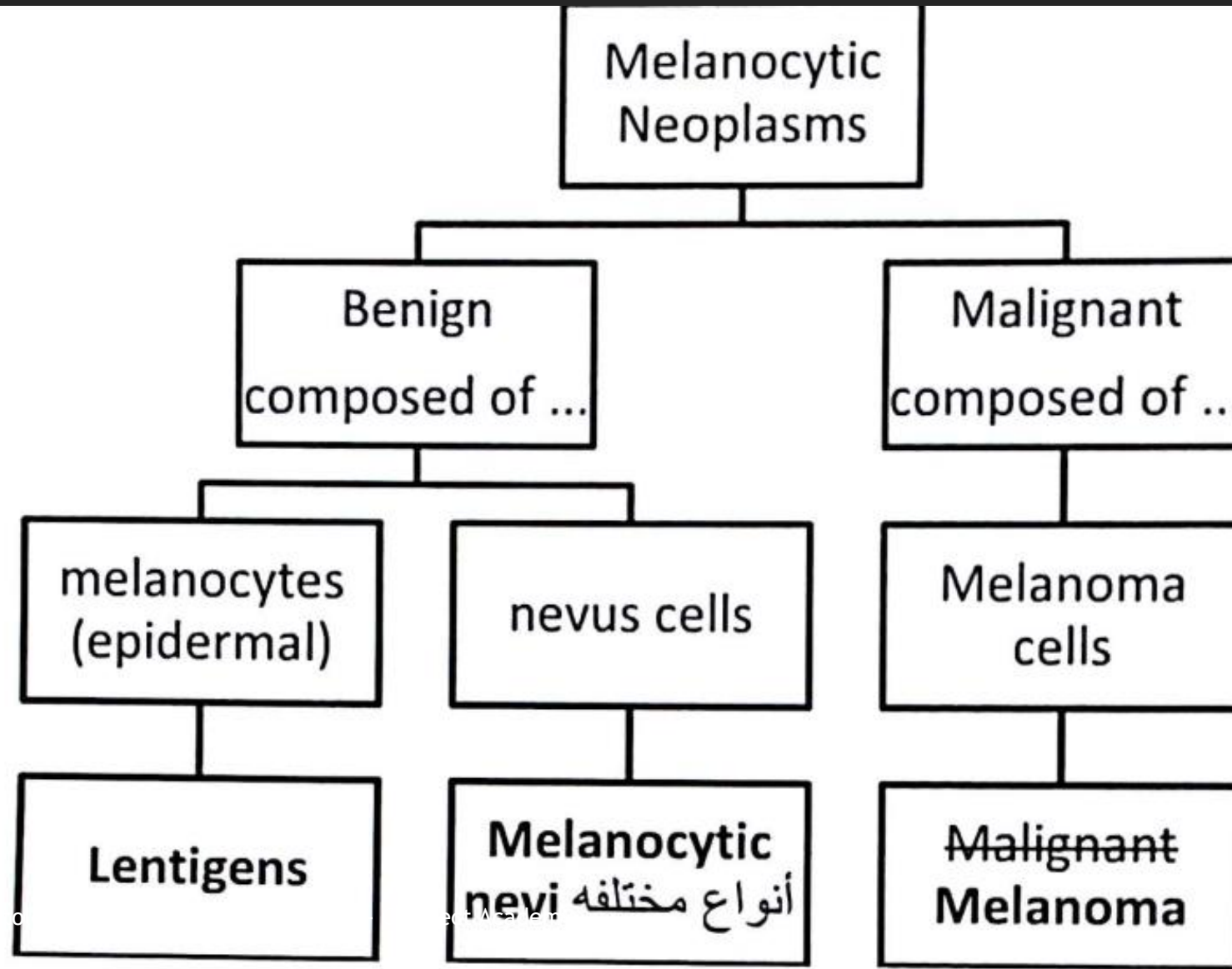


Benign Melanocytic Neoplasms

1

Dr eman gomaa
Consultant dermatologist



Melanocytes	Nevus Cells	Melanoma Cells
Cytoplasm is dendritic	Rounded or spindle	Rounded or spindle
Cells are solitary	Arranged in clusters	In clusters and large sheets
Nuclei are small and regular	Nuclei of most cells are small and regular	Most nuclei are large, irregular, and hyperchromatic
Mitoses are rare	Mitoses are rare	Mitoses are usually present

Ephelides	
Café-au-lait macules	
Becker melanosis (nevus)	
Solar lentigines.	
Lentigo simplex and mucosal melanotic lesions	
Dermal melanocytosis	
Nevus of Ota and related conditions	
Blue nevus and its variants	
Common acquired melanocytic nevi	
Melanocytic nevi of genital skin and other “special sites”	
Melanocytic nevus of acral skin	
Spitz (spindle and epithelioid cell) nevus/tumor	
Pigmented spindle cell nevus	
Atypical (dysplastic) melanocytic nevus	
Congenital melanocytic nevus	
Nevus spilus.	
Halo nevus	
Combined nevus.	
Recurrent melanocytic nevus.	

EPHELIDES

5

Small, well-circumscribed, pigmented macules found only on sun-exposed skin of individuals with fair skin

Clinical Features

Ephelides only occur on sun-exposed areas of the body, in particular the face, dorsal aspects of the arms, and upper trunk. They do not occur on mucous membranes. Ephelides are well demarcated and are round, oval, or irregular in shape. They are usually 1 to 3 mm in diameter, but are occasionally larger. Depending on the intensity of sun exposure, ephelides vary in color from light to dark brown, but almost never become as dark as lentigines or junctional nevi. They may increase in



COMPARATIVE CLINICAL FEATURES OF EPHELIDES AND SOLAR LENTIGINES		
	Ephelides (freckles)	Solar lentigines (see Ch. 109)
Epidemiology		
Age of appearance	Early childhood	By 20–30 years (I, II)
Skin color	Light pigmentation	Light to dark pigmentation
Hair color	Often red or blond hair	Any type
Skin phototype	More common in I, II	More common in I and II, but also in III and IV
History		
Precipitating factors	Bursts of high-intensity sun exposure activate latent predetermined melanocytes, leading to permanent freckles	Repeated sun exposure over time alters melanocytes within a circumscribed area to overproduce melanin
Duration of lesions	Fade without sun exposure	Persist for life
Relation to season	Much darker in summer, fade in winter and over time with aging	May darken in summer but do not fade in winter
Heredity	Probably AD; also AR when related to <i>MC1R</i> variants	No data
Physical examination		
Type of lesion	Macule	Macule
Size	1–5 mm	5–15 mm or larger
Color	Light or medium brown	Medium or dark brown
Shape	Oval to irregular	Oval to irregular
Border	Smooth to jagged	Smooth to jagged
Distribution	Favor the face, extensor forearms, and back; rare on the dorsal aspect of the hands	Sites of chronic sun exposure, especially the face, arms (including dorsal aspect of the hands), and upper trunk
Dermoscopic features		
	Uniform pigmentation and a moth-eaten edge	Diffuse light brown structureless areas, sharply demarcated and/or moth-eaten borders, fingerprinting, and a reticular pattern with thin lines that are occasionally short and interrupted

- **Solar lentigo**

- Multiple pigmented macules on sun-exposed areas; most common in Caucasians (essentially ubiquitous after 60 years of age) > light-skinned Asians
- Histology: elongated bulbous rete ridges with hyperpigmentation (“**dirty socks**”); +/- mild increase melanocyte density; solar elastosis in dermis



Fig. 112.3 Solar lentigines. Numerous light brown macules, some of which have an irregular border, on chronically sun-exposed skin.

- **Lentigo simplex**
 - Well-demarcated, evenly pigmented brown to black macule; any age and any anatomic site
 - Histology: basal layer hyperpigmentation; elongated rete ridges with mild ↑ melanocyte density
 - Conditions a/w multiple lentiginos:
 - **LEOPARD**
 - **Carney complex (LAMB/NAME)**
 - **Peutz-Jeghers (especially oral/perioral)**
 - Laugier-Hunziker
 - Cowden syndrome
 - **Bannayan-Riley-Ruvalcaba (penile)**
 - Xeroderma pigmentosum
 - Cronkhite-Canada



Fig. 112.4 Genital lentiginosis. Biopsy of the clinically most atypical lesion (at 7 o'clock) showed no cellular atypia. Over a period of 10 years, several of the macules faded. *Courtesy, Jean L. Bolognia, MD.*

DISORDERS ASSOCIATED WITH MULTIPLE LENTIGINES			
Disorder	Comments	Disorder	Comments
Generalized		Localized	
LEOPARD syndrome (now referred to as Noonan syndrome with multiple lentigines)	• Autosomal dominant inheritance (high penetrance, variable expressivity); mutations in <i>PTPN11</i> > <i>RAF1</i> > <i>BRAF</i> • Lentigines present in infancy/early childhood • Café-noir macules • ECG changes (conduction defects, hypertrophic cardiomyopathy), ocular hypertelorism, pulmonary stenosis, abnormal genitalia, growth retardation and deafness	<i>Head and neck (including oral mucosa) ± acral</i>	
		Peutz-Jeghers syndrome	• Autosomal dominant inheritance; mutations in <i>STK11</i> • Lentigines favor perioral region[†], oral mucosa[‡] and hands; longitudinal melanonychia • Multiple hamartomatous GI polyps • Pancreatic carcinoma; ovarian (adenoma malignum)/testicular tumors
Generalized lentigines	• Autosomal dominant inheritance; mapped to 4q21.1–q22.3 • Café-au-lait macules	Bandler syndrome [*]	• Autosomal dominant inheritance • Distribution of lentigines similar to Peutz-Jeghers • GI bleeding with small intestine hemangiomas rather than polyps
Deafness plus lentiginosis [*]	• ? Forme fruste of LEOPARD syndrome		

DISORDERS ASSOCIATED WITH MULTIPLE LENTIGINES

Disorder	Comments	Disorder	Comments
Carney complex (NAME/LAMB syndrome)	• Autosomal dominant inheritance; mutations in <i>PRKAR1A</i> (second locus on 2p16) • Lentigines, mucocutaneous myxomas, blue nevi (including epithelioid) • Psammomatous melanotic schwannomas • Atrial myxoma • Myxoid mammary fibroadenomas • Pigmented nodular adrenocortical disease; testicular (calcifying Sertoli cell), thyroid, and pituitary tumors	Laugier–Hunziker syndrome	• Similar distribution of lentigines as in Peutz–Jeghers, including lips, oral mucosa and digits • Longitudinal melanonychia and genital melanosis
		Hyperkeratosis-hyperpigmentation (Cantú) syndrome	• Autosomal dominant inheritance • Punctate palmoplantar keratoderma • Multiple small (1 mm) macules on the face, forearms, hands/feet
Arterial dissection plus lentiginosis	• ? Inheritance pattern • Cutaneous lentigines with onset in childhood • Dissection of aortic, internal carotid and vertebral arteries	Cowden disease	• Autosomal dominant inheritance; mutations in <i>PTEN</i> • Periorificial and acral pigmented macules (see Table 63.3)
Gastrocutaneous syndrome*	• Autosomal dominant inheritance; see Table 61.4 for details	Centrofacial lentiginosis (neurodysraphic)	• Autosomal dominant inheritance • Lentigines in a butterfly distribution on the nose, cheeks > forehead, eyelids, upper lip • Onset in infancy; increase in number in childhood • Possibly associated with neuropsychiatric illness and osseous anomalies
Tay syndrome*	• ? Autosomal recessive inheritance • Growth retardation, intellectual disability; triangular face; cirrhosis; trident hands • Café-au-lait macules; premature canities; vitiligo		
Pipkin syndrome*	• Autosomal dominant inheritance • Nystagmus; strabismus	Inherited patterned lentiginosis	• Autosomal dominant inheritance • African-Americans with light brown skin • Pigmented macules appear in early childhood • Present on central face and lips > buttocks, elbows, hands/feet • Rare oral mucosal involvement
		Cronkhite–Canada syndrome	• Typically affects older men • Lentigines of buccal mucosa, face, hands/feet • Alopecia (diffuse, non-scarring), nail dystrophy, intestinal polyposis, diarrhea, malabsorption



Fig. 112.5 Labial and acral lentigines in two patients with Laugier-Hunziker syndrome.

A Multiple hyperpigmented macules on the lower lip and a few on the tongue; Peutz-Jeghers syndrome can have a similar clinical appearance.

B Numerous lentigines on the palms. *B, Courtesy, Drs Tello Flores and Sánchez Félix.*

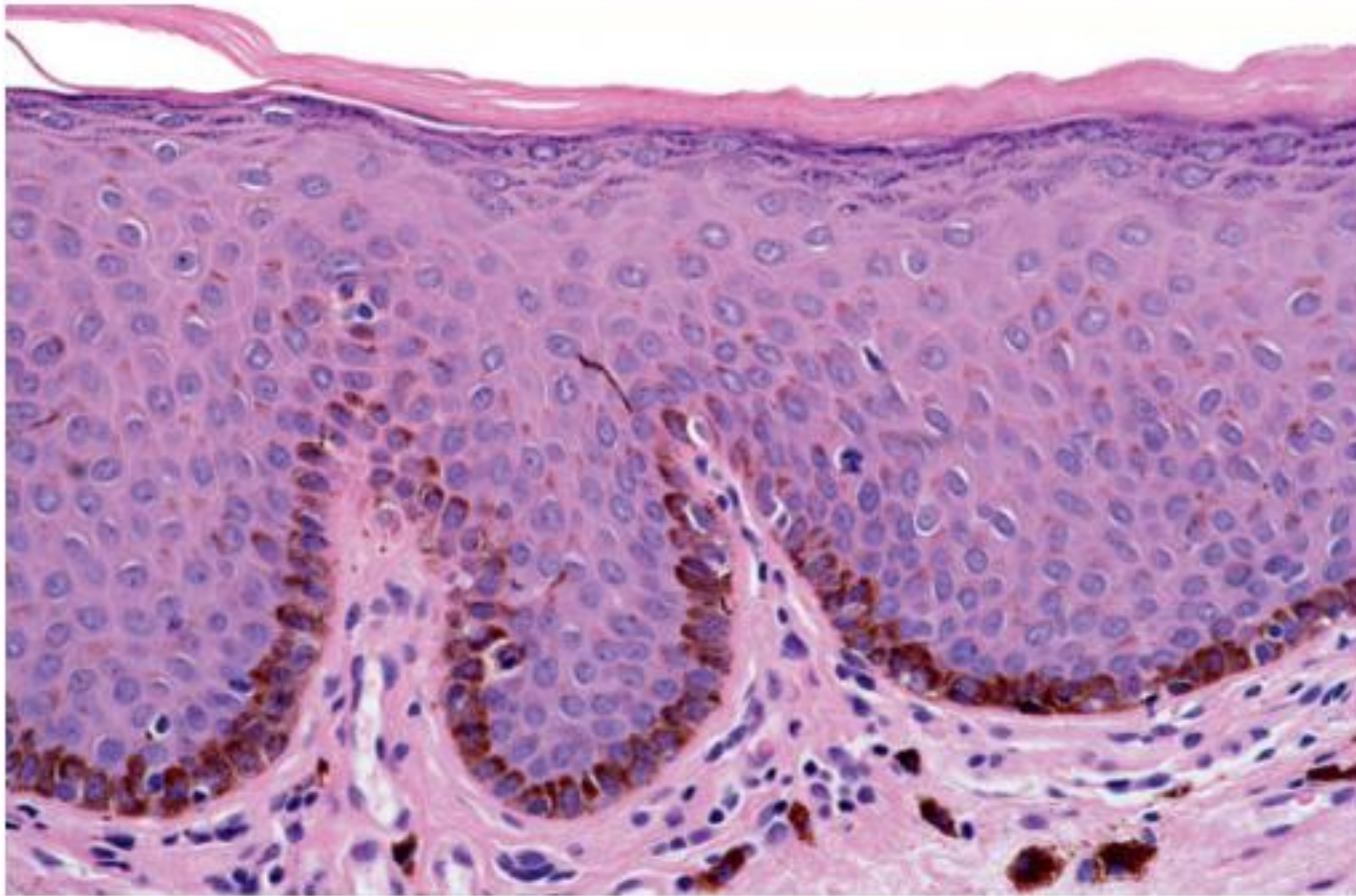


Fig. 112.6 Labial melanotic macule (lentigo) – histopathologic features. Hyperpigmented rete ridges are broader than in the typical lentigo. In addition to basilar hyperpigmentation and melanophages, a few dendritic processes of melanocytes are seen. *Courtesy, Lorenzo Cerroni, MD.*

- **Becker's melanosis (NOT a true melanocytic disorder)**
 - 0.5% prevalence, more common in adolescent and **young adult males**
 - May be as a result of androgen-mediated hyperplasia
 - Onset around **puberty**
 - Hyperpigmented plaque with thickening, irregularity, and/or hypertrichosis on **upper torso** most commonly **ipsilateral breast hypoplasia** seen in some patients (Poland syndrome), rarely associated with skeletal defects and/or limb asymmetry



Fig. 112.2 Becker melanosis (nevus). **A** Unilateral hyperpigmented patch of the upper chest with confluence centrally and an irregular broken-up border. Acneiform lesions were limited to the lesion in this prepubescent boy. **B** Unilateral, block-like configuration with hyperpigmentation and hypertrichosis; the patient had an underlying smooth muscle hamartoma. **C** Large patch of hyperpigmentation on the leg which is medium brown in color. Such lesions may be misdiagnosed as café-au-lait macules or congenital melanocytic nevi, especially when they do not occur on the upper trunk. A, Courtesy, Julie V Schaffer, MD; B, Courtesy, Edward Cowen, MD; C, Courtesy, Jean L Bolognia, MD.



Fig. 112.1 Café-au-lait macules (CALMs).
A Typical CALM with well-defined and regular margins in a patient with neurofibromatosis 1.
B Large tan patch in a geographic pattern on the lateral trunk. The patient did not have McCune–Albright syndrome. *A, Courtesy, Julie V Schaffer, MD.*



1. Benign melanocytic neoplasms

Melanocytic nevi

Types:

- a. Common naevi
- b. Special naevi
- c. Dermal naevi

- **Dermal melanocytosis**
 - Congenital (Mongolian spot): present at birth in most **Asians and blacks**; **lumbosacral** region; presents with (p/w) gray-blue patch (as a result of the **Tyndall effect** where shorter light wavelengths are reflected by melanocytes); often **resolves** during childhood
 - Histology: sparsely distributed elongated dendritic melanocytes in lower 2/3 dermis, lying parallel to epidermis

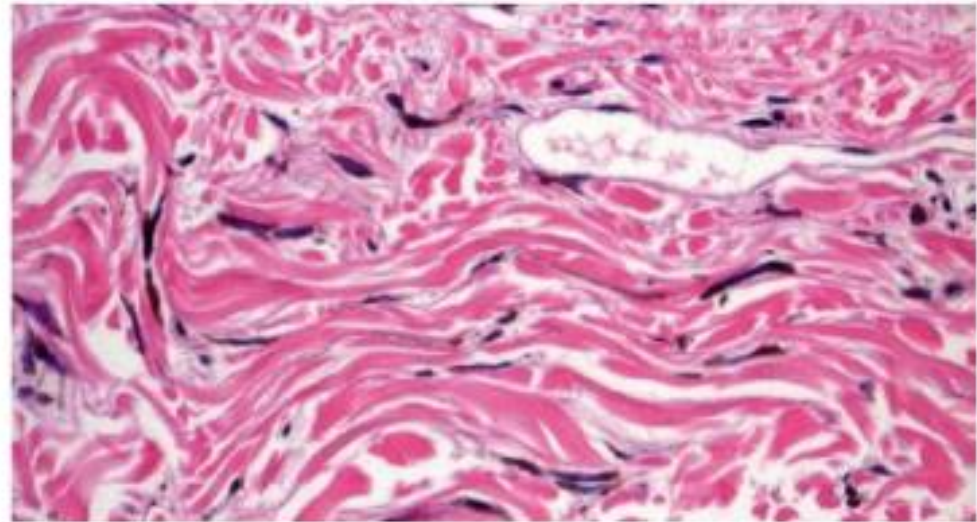


Fig. 112.7 Dermal melanocytosis (Mongolian spots) in a child with neurofibromatosis 1. Surrounding each café-au-lait macule there is an absence of the characteristic blue discoloration.

- **Nevus of Ota:**
- presents in first year of life or around puberty; ↑incidence in pigmented individuals (Asians and blacks); p/w coalescing gray/blue macules in **V1/V2 distribution** and frequent **scleral involvement** (60%); unilateral (90%) > bilateral; persists for life; may **enlarge under hormonal influences**; **10% develop glaucoma**; rare malignant degeneration to **uveal melanoma** (perhaps higher risk in Nevus of Ota lesions with activating mutations in *GNAQ*)
 - **Histology:** elongated dendritic melanocytes more numerous than in congenital dermal melanocytosis; involves upper dermis
- Other clinical variants:
 - **Nevus of Ito:** located on the shoulder, supraclavicular, and scapular regions; essentially no risk of progression to melanoma
 - **Hori's nevus:** acquired nevus of Ota-like macules **bilateral** zygomatic region; East Asian females
 - **Sun's nevus:** acquired, **unilateral** variant of Hori's nevus



Fig. 112.8 Nevus of Ota (oculodermal melanocytosis) – clinical and histopathologic features. A Unilateral blue-gray discoloration of the face, which is either mottled or confluent. There is also involvement of the sclera. **B** Scattered pigmented elongated and dendritic melanocytes within the reticular dermis. *R. Courtesy, Lorenzo Cerroni, MD.*





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eFig. 112.4 Nevus of Ota – ocular involvement. Blue discoloration of the sclera. *Courtesy, Jean L. Bolognia, MD.*

- Variants:
 - Common blue nevus:

- - Cellular blue nevus:
 - Blue/gray/black plaques or nodules; often **larger** (1–3 cm); favor **buttocks or scalp**
 - **Dense proliferation of plump/fusiform pale gray melanocytes** (“**dumbbell configuration**”)
- - Epithelioid blue nevus
- - Malignant blue nevus (= melanoma): often arises **within cellular blue nevus**;

- **Blue nevus**

- Onset in childhood/adolescence, but can also occur in older patients, 25% of cellular blue nevi are congenital
- Most common sites: **scalp, sacral area, and distal extensor extremities**
- Derived from dermal melanocytes (persist during embryogenesis rather than populating epidermis)
- Activating mutations in **GNAQ** and **GNA11** seen in 83%; results in downstream MAPK pathway activation
- Same mutations are the most common mutations in **uveal melanoma** (46%; concomitant *BAP-1* loss in uveal melanoma leads to increased risk of metastasis and death)



Fig. 112.9 Common blue nevus – clinical and histopathologic features. **A** Well-circumscribed, dark blue flat papule. **B** Dark blue-black papule in a patient with more darkly pigmented skin. **C** Blue papule with an irregular, micropapular surface. **D,E** Heavily pigmented, spindle-shaped melanocytes as well as melanophages within the dermis; the melanocytes are denser than in dermal melanocytosis or nevus of Ota. *A–C, Courtesy, Julie V Schaffer, MD; D,E, Courtesy, Lorenzo Cerroni, MD.*

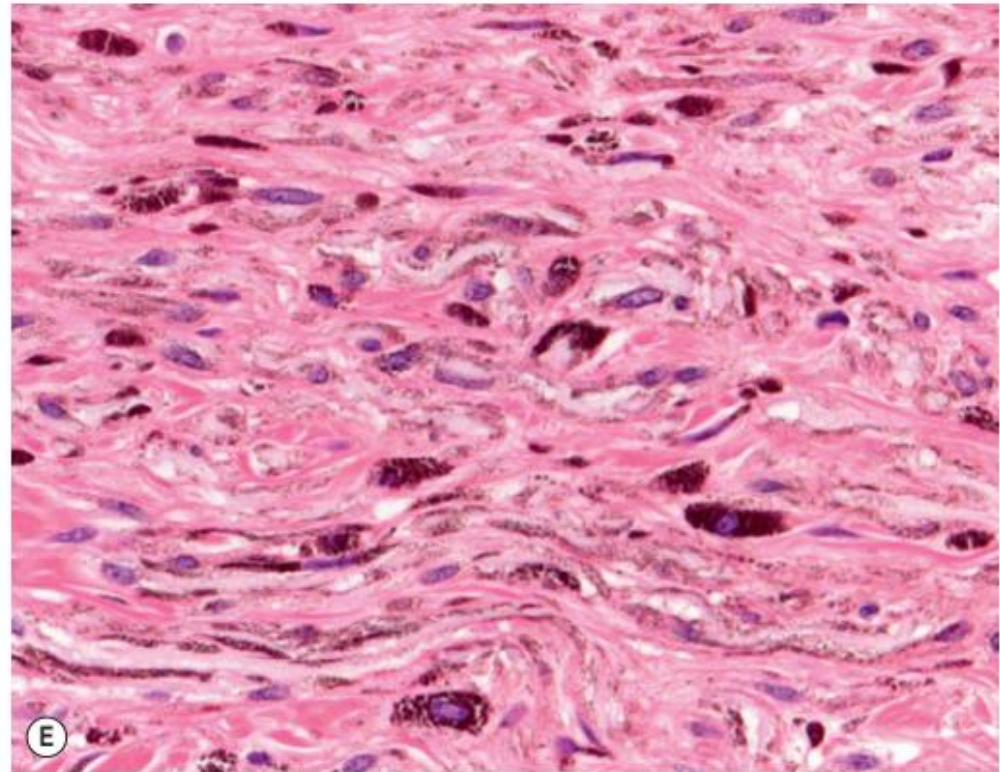
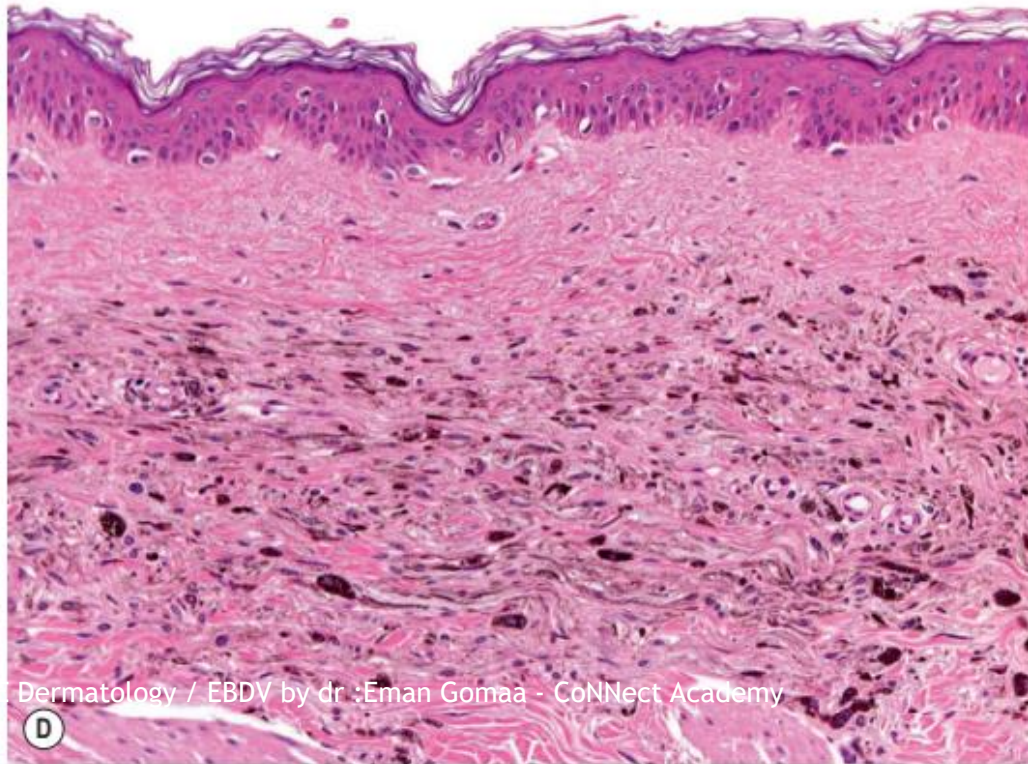
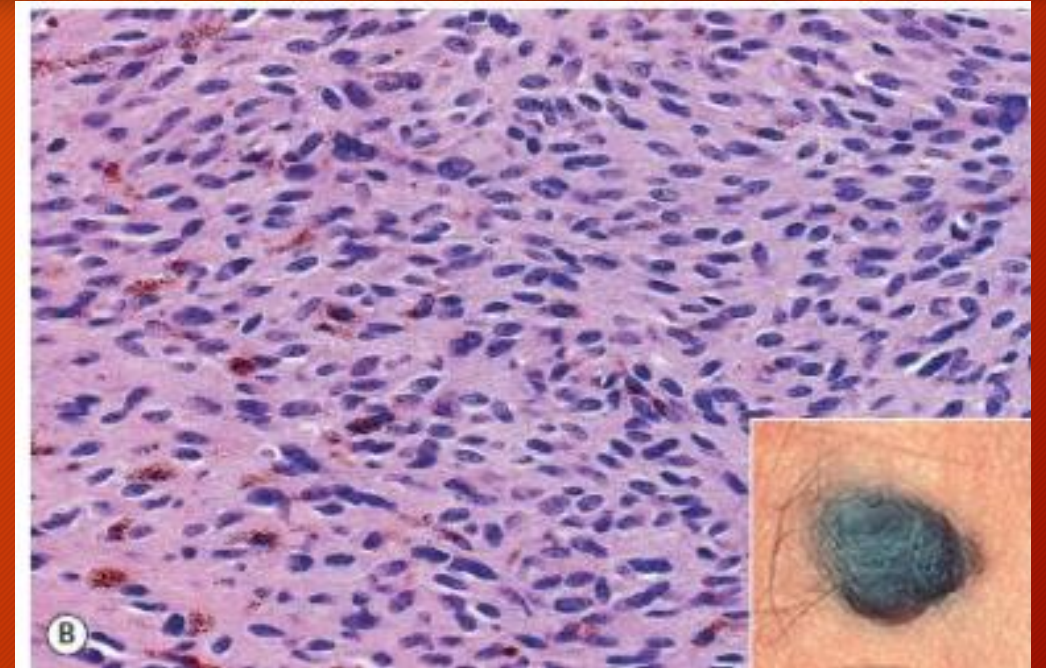




Fig. 112.10 Cellular blue nevi – clinical and histopathologic features. A A black nodule is seen. **B** Plump spindled melanocytes devoid of melanin pigment and arranged within bundles are present within the dermis. Scattered melanin-containing melanophages are also present. A firm blue plaque is a common presentation (inset). A,B, Courtesy, Lorenzo Cerroni, MD; Inset, Courtesy, Jean I. Bolognia, MD.



- **Halo nevus (Sutton nevus)**

- Pigmented nevus with surrounding hypopigmented zone; most commonly second decade; most commonly on the **back**; most commonly benign
 - May be a/w vitiligo or melanoma (rarely) at another site
- Multiple lesions can occur idiopathically or w/ infliximab
- Histology: bland nevus w/ **lymphocytes intertwined (“mingling”)** with **melanocytes**
 - In contrast, lymphocytes form a lichenoid band (“**riot police barrier**”) under and around melanoma



Fig. 112.25 Halo melanocytic nevi. **A** Stage I (central brown nevus), II (central pink nevus), III (depigmented macule), and early IV (partially repigmented skin) halo nevi are present. Multiple halo nevi are seen most commonly in children and young adults with numerous nevi. **B** A benign compound nevus with an asymmetric halo.
B, Courtesy, Jean L Bolognia, MD.

- **Spitz nevus**

- Acquired, usually solitary lesions in **first two decades** (use caution in diagnosing a patient in the fourth decade and older); most common on **head/neck** > extremities

- Pathogenesis:

- **HRAS** mutations/**11p gains**

- No *BRAF* mutations

Rapidly growing pink-red papulonodule; usually <1 cm

Kamino bodies: pink clumps of **BMZ material**

(**collagen IV**) within epidermis



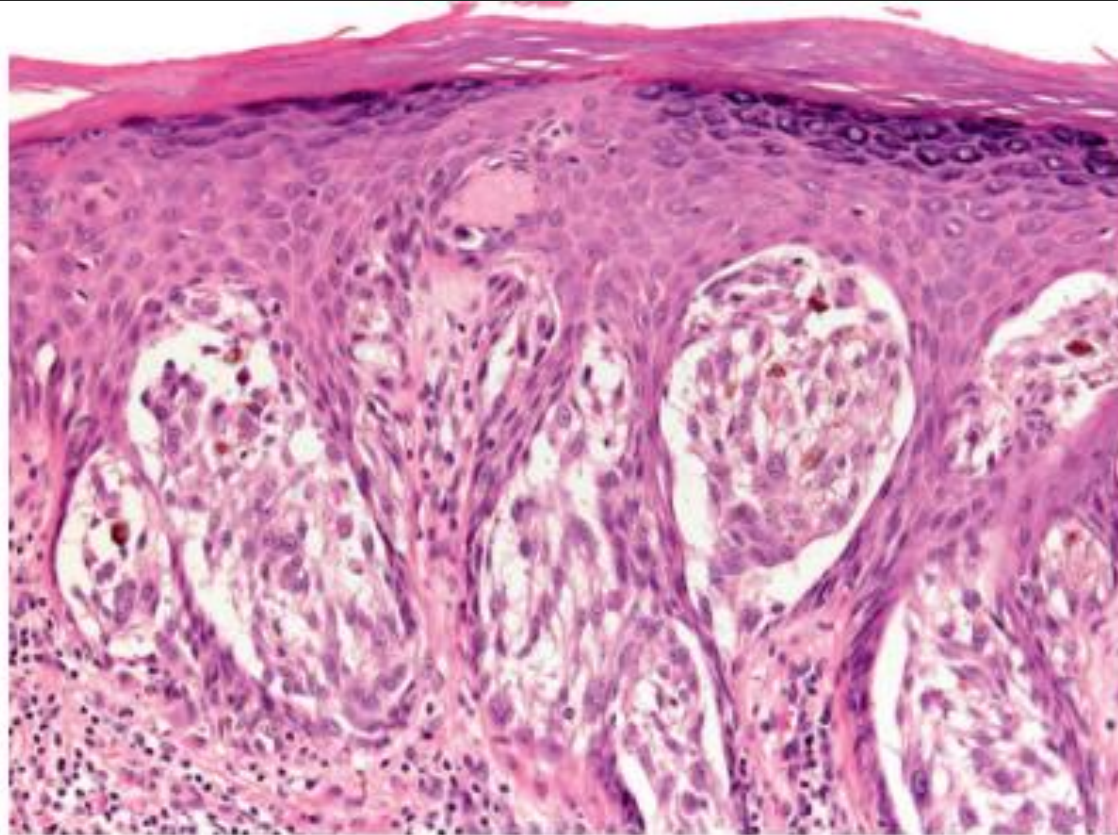


Figure 6-5. Histology of Spitz nevus. Nests of cohesive spindled and epithelioid melanocytes with clefts between the nests and the hyperplastic epidermis.

- **Pigmented spindle cell nevus of Reed**
 - Heavily pigmented variant of Spitz nevus **comprised ~exclusively of spindled spitzoid melanocytes**
 - **Young F > M; thigh** most commonly (> other extremities and trunk)
 - Darkly pigmented macule/papule, usually <6 mm
 - Histology: **junctional or superficial compound**, symmetric, circumscribed proliferation of **spindled melanocytes** arranged in vertically oriented fascicles + **numerous melanophages** (much more than in conventional Spitz nevi) in superficial dermis +/- **pigmented parakeratosis**
 - Architecture: similar to Spitz nevus but almost always **confined to junction** (or very superficial dermis)
 - Cytology: spindled melanocytes are same as in Spitz nevi, but **PSCN lacks the epithelioid cells** seen in conventional Spitz nevi

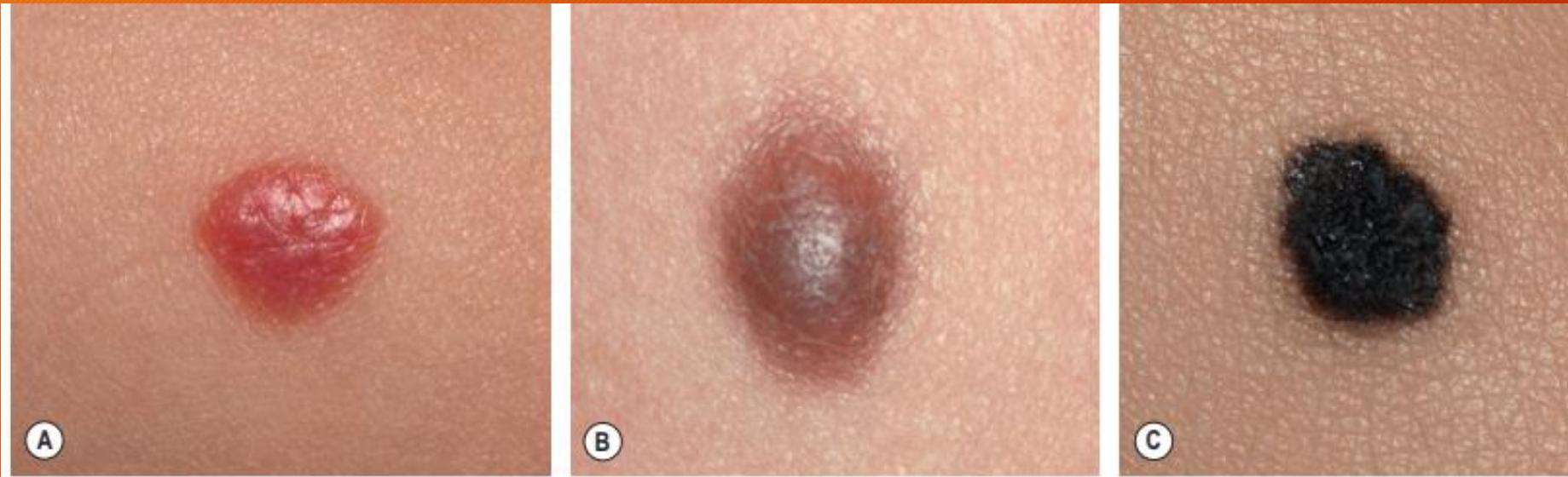


Fig. 112.15 Spitz nevi. **A** Red, symmetric, dome-shaped papule on the thigh of a child. **B** Medium-brown, symmetric, dome-shaped papule with a thinner rim in an adolescent. **C** Darkly pigmented Spitz nevus in a child with a flatter surface. *A,C, Courtesy, Julie V Schaffer, MD; B, Courtesy, Lorenzo Cerroni, MD.*

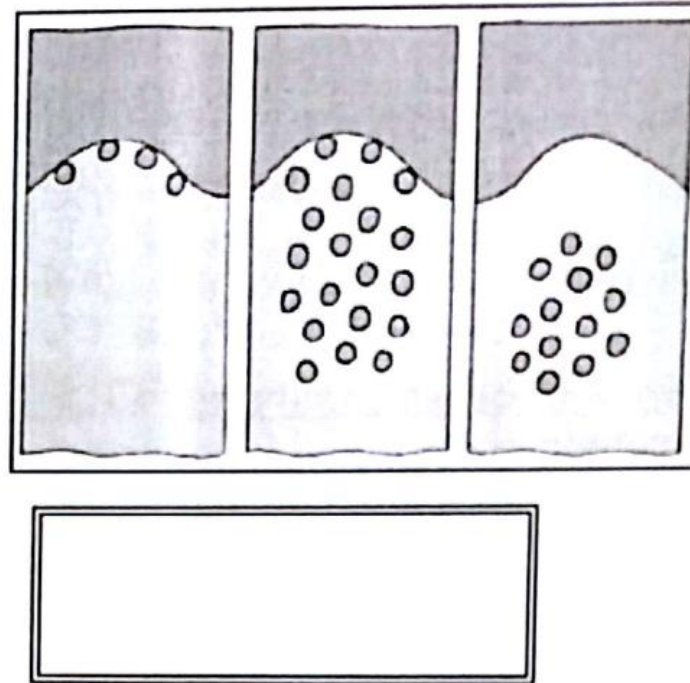
a. Common naevi

(Acquired melanocytic naevi, nevo-cellular nevi, moles)

Pathogenesis:

Proliferation of melanocytes at the dermal-epidermal junction $\Rightarrow$ clusters (**neests**) of melanocytes (*naevus cells*) $\Rightarrow$

1. all cells remain attached to the basal layer of the epidermis $\Rightarrow$ **junctional** naevus
2. some cells may migrate into the dermis $\Rightarrow$ **compound** naevus
3. **The end stage:** no naevus cells are attached to the epidermis + all cells are lying free in the dermis $\Rightarrow$ (mature) **intradermal** naevus



TYPES OF MOLES

CONGENITAL MELANOCYTIC NEVI



Combined
Nevus



Compound
Nevus

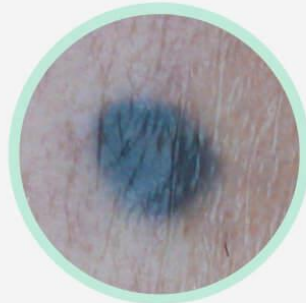


Junctional
Nevus



Intradermal
Nevus

ACQUIRED MELANOCYTIC NEVI

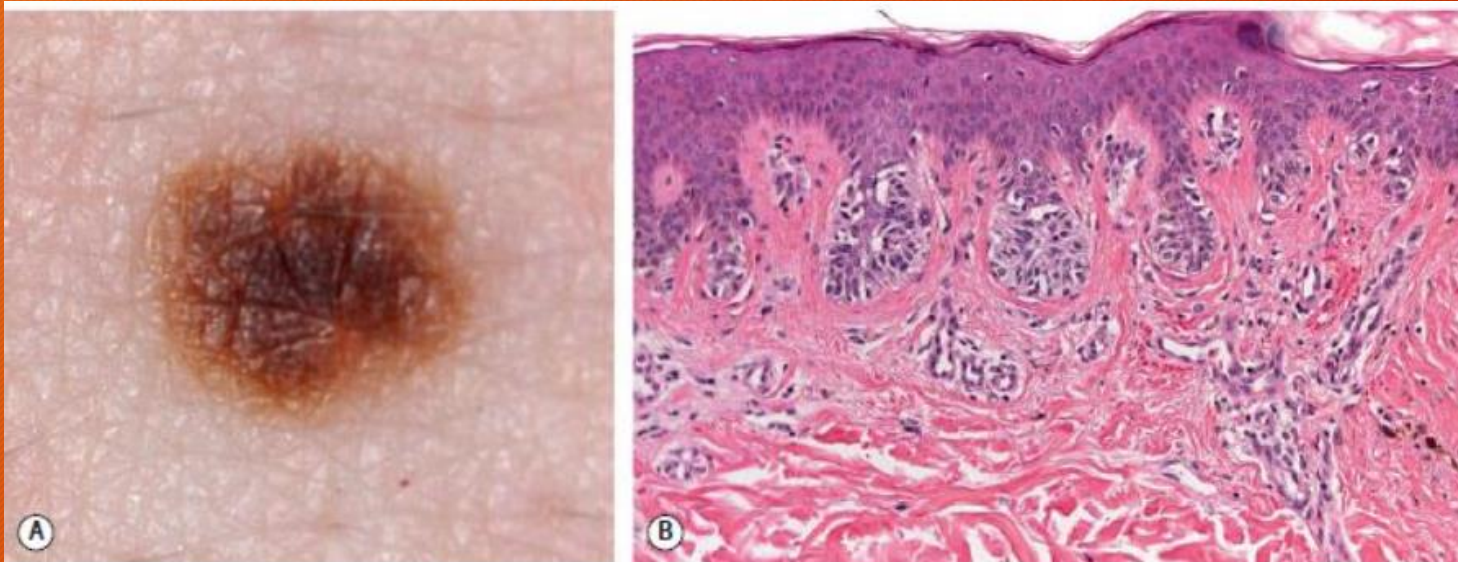


Blue Nevus

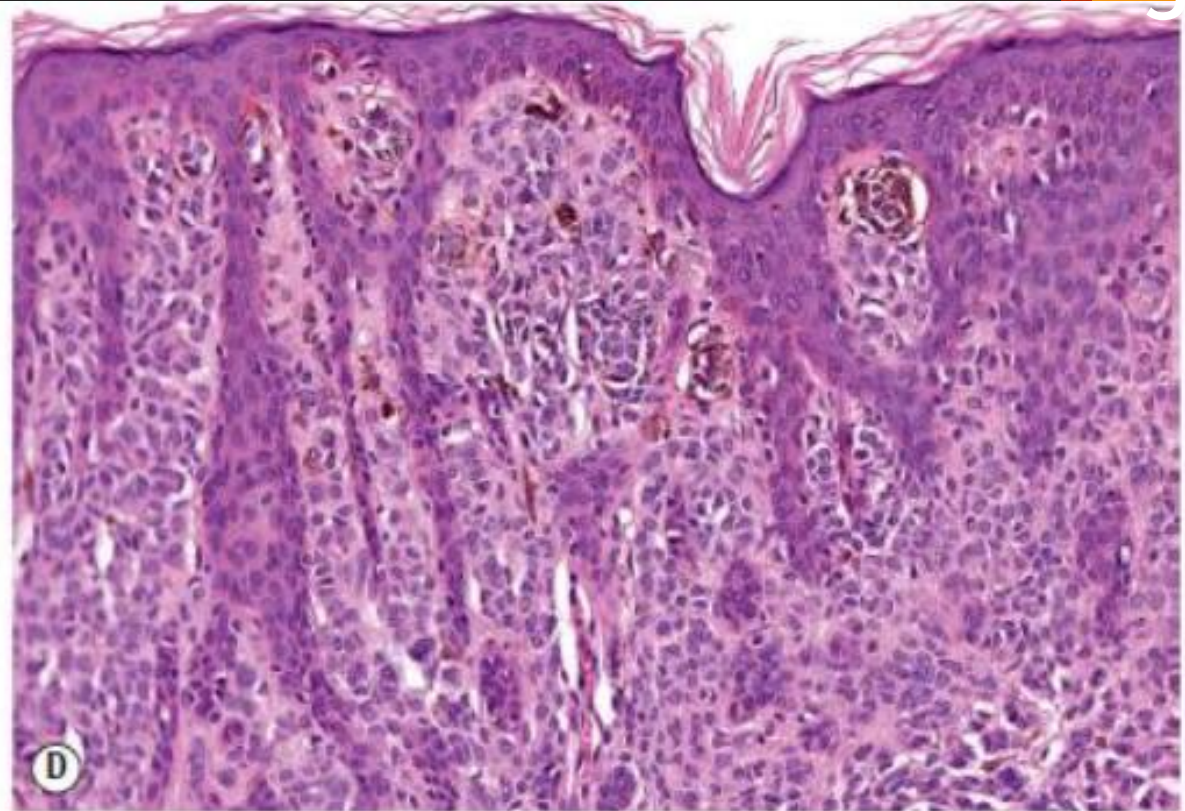


Dark Nevus

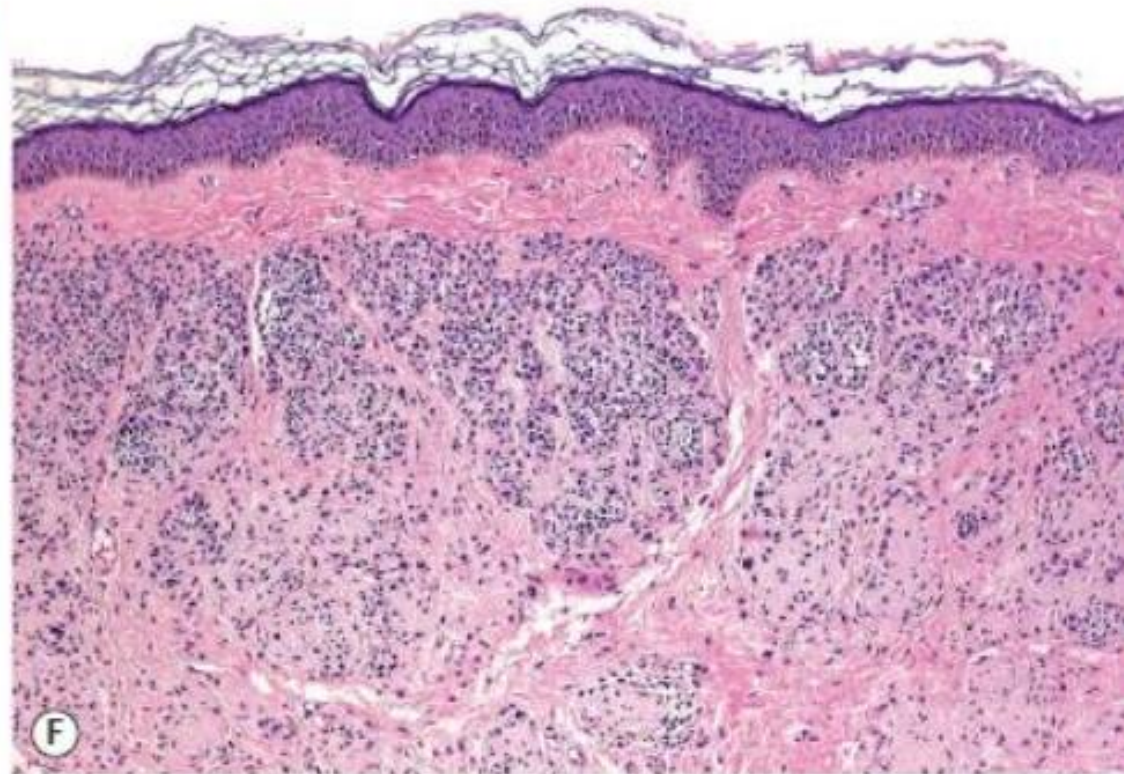
	Junctional	Compound	Intradermal
Clinical:	Pigmented Not raised (flat)	Pigmented (epidermal component) Raised (dermal component)	Non-pigmented Raised (dermal component)
Histopathology:	Nests of melanocytes at the <i>dermo-epidermal junction</i> only	Nests of melanocytes at the <i>dermo-epidermal junction</i> and in the <i>dermis</i>	Nests of melanocytes in the <i>dermis</i> only
		The dermal cells: • The more superficial cells (type A, epithelioid) as naevus cells, form melanin • The deeper cells (type B, lymphocytoid) are smaller and usually contain no melanin – arranged as a band • In the deeper dermis (type C, spindle): cells are arranged as arborizing columns, spindle-shaped (resemble cells of neural or fibroblast lineage – WHY neural??)	



- Junctional nevus – dark brown macule with lighter brown rim and regular nests of small, uniform nevomelanocytes at the dermal–epidermal junction.



- Compound nevus – light to medium brown papule and both junctional and dermal nests of nevus cells.



- Intradermal nevus – soft light pink papule and nests of nevus cells within the dermis

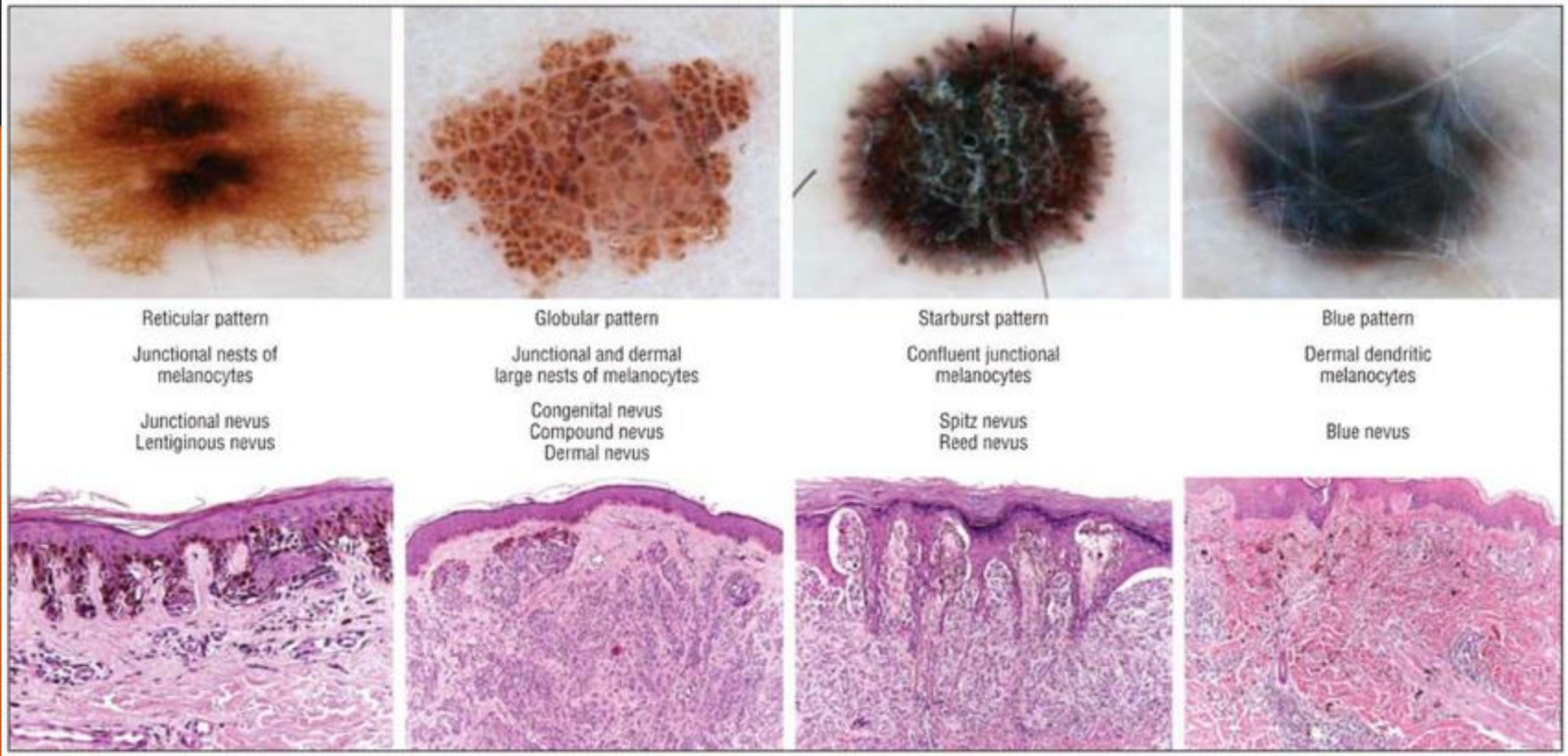


Fig. 112.14 Acral melanocytic nevus. The parallel furrow pattern is evident with a small focus of the lattice pattern. The ostia of the eccrine ducts within the epidermal ridges can also be seen, especially at the site of blue ink.

Courtesy, Jean L. Bolognia, MD.

ACRAL NEVI – COMMON DERMOSCOPIC PATTERNS

Pattern	Dermoscopy features
Parallel furrow (most common)	Parallel rows of linear pigmentation along furrows/valleys (sulci of the surface skin markings; see Figs 112.13 & 112.14)
Lattice-like	Parallel furrows plus lines of pigmentation crossing the parallel lines (see Fig. P4)
Fibrillar	Densely packed, fine parallel lines of pigmentation, often set in a direction that crosses the skin markings



Atypical (dysplastic) melanocytic nevus

- Occur sporadically or in the setting of familial atypical multiple mole melanoma syndrome (FAMMM)
 - FAMMM: AD inheritance; characterized by multiple melanocytic nevi (50+), family history of melanoma, and mutations in **CDKN2A** gene (encodes p16 and p14_{ARF})
- Any site (**trunk and scalp** most common)



Fig. 112.20 Atypical melanocytic nevus.
A There is asymmetry as well as several shades of brown, simulating the clinical features seen in cutaneous melanoma.
B In addition to multiple atypical nevi, patients can have numerous banal nevi.



- ***NRAS* mutations** in majority of cases ($> \textit{BRAF}$ mutations)
 - Divided into small (<1.5 cm), medium (1.5 cm–19.9 cm), and **large (>20 cm)**



Fig. 112.22 Congenital melanocytic nevi associated with neurocutaneous melanosis. A Multiple medium-sized nevi. The patient had 25–30 such lesions. This presentation is most commonly associated with neurocutaneous melanosis. **B** Giant congenital nevus in an axial location with numerous satellite nevi, some of which have hypertrichosis. This patient died of intractable ascites due to the migration of melanocytes from the brain to the peritoneal cavity via his VP shunt. In the perianal region, the nevus was softer and somewhat “spongy” due to neurotization. *Courtesy, Jean I. Bolognia, MD.*



Nevus spilus/speckled lentiginous nevus

40

- Presents within the first year of life
- Homogeneous tan patch within which develops small pigmented macules and papules
- Trunk and extremities most commonly affected sites
- May be associated with **phakomatosis**



GENES IMPLICATED IN THE DEVELOPMENT OF MELANOCYTIC NEVI

Gene	Type of melanocytic nevus
<i>BRAF</i>	Common acquired melanocytic nevi (65–80%)
<i>NRAS</i>	Congenital melanocytic nevi (~80%), small minority of common acquired melanocytic nevi, congenital melanocytic nevus with hypophosphatemia (single case)
<i>HRAS</i>	Spitz nevi ^{*,**} (~15%), deep penetrating nevi (6%), nevus spilus/speckled lentiginous nevi (100% in one series); also phakomatosis pigmentokeratolica
<i>GNAQ</i>	Blue nevi (50–85%, depending upon the study); also phakomatosis pigmentovascularis
<i>GNA11</i>	Blue nevi (7–10%); also phakomatosis pigmentovascularis
<i>BAP1</i>	Atypical epithelioid cell nevi/tumors (often polypoid and combined dermal nevi with large “spitzoid” cells)
<i>PTEN</i>	Nevi in patients with xeroderma pigmentosum (~60%)
<i>PRKAR1A</i>	Epithelioid blue nevi (mutation in the second allele in Carney complex), pigmented epithelioid melanocytomas

GENETIC SYNDROMES ASSOCIATED WITH NUMEROUS ACQUIRED MELANOCYTIC NEVI

Familial atypical mole and melanoma syndrome*

BAP1 cancer syndrome – atypical epithelioid nevi/tumors, uveal melanomas, mesothelioma

Carney complex (NAME/LAMB syndromes) – see [Table 112.2](#)

Turner syndrome – webbed neck, congenital lymphedema (partial or complete loss of X chromosome in females)

Noonan syndrome (mutations in multiple genes including *PTPN11*, *RAF1*, *KRAS*, *SOS1*) > other RASopathies (e.g. LEOPARD [Noonan with multiple lentigines] syndrome, cardiofaciocutaneous syndrome, Costello syndrome; see Ch. 55)

Ectrodactyly, ectodermal dysplasia, and cleft lip/palate syndrome (EEC syndrome; *TP63* mutations)

Goeminne syndrome – torticollis, keloids, cryptorchidism, and renal dysplasia

Kuskokwim syndrome – congenital joint contractures, skeletal deformities (*FKBP10* mutations)

Mulvihill Smith syndrome – progeroid short stature with pigmented nevi

Trichorhinophalangeal syndrome type II (Langer-Giedion syndrome)

Tricho-odonto-onychial dysplasia

